

- 7) What is the message from the Kolm's index? Calculate the Kolm index for a distribution, which frequency 5 for value Rs.5 lakh and frequency 5 with value Rs.10 lakh and therefore total size 10 and the arithmetic mean 7.5.

.....

.....

.....

.....

.....

11.3 GINI INDEX

This coefficient of concentration, as it is usually called, owes to an Italian statistician by the name of Corrado Gini (1912). Modern practice is to call it just Gini. This index in its origin is positive. There are a number of ways in which this coefficient can be expressed. There are also a number of ways in which it can be interpreted. People have also derived it as a measure of inequality under plausible axioms in welfare theoretic framework. The index satisfies many axioms proposed in literature for an index of inequality.

First, we shall discuss those definitions and expressions, which can be derived as a measure of dispersion. Besides giving its expressions for its frequency data for grouped observations, we shall discuss its welfare theoretic interpretations.

11.3.1 Gini as a Measure of Dispersion

Recall that mean deviation and standard deviation, which are measures of dispersion, seek the deviation from arithmetic mean. Also recall that one of the logarithmic measure sought deviation from the geometric mean. However, one may ask why to seek dispersion in terms of deviations from any mean? Why not compare all pairs and seek the differences. In order to consider positive values of differences, we either take modal values of deviations before averaging (mean deviation) or sum the squares of deviations and take root of the mean of the squared differences.

Corrado Gini (1912) proposed to consider all the differences, that is all pairs of values. By contrast, the range measure of dispersion considers only one pair of highest value and lowest value. When x_i and x_j denote i th and j th incomes respectively and $i, j=1, 2, \dots, N$, we can see that the aggregate of absolute differences is given by

$$\sum_{i=1}^N \sum_{j=1}^N |x_i - x_j| \quad (G.1)$$

and because total number of differences is N^2 , the mean of absolute differences can obviously be written as

$$\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N |x_i - x_j| \quad (G.2)$$

where the differences with the self have also been counted and the difference of x_i with x_j is treated as separate from that of x_j and x_i though numerically they are the same. This is also said to be the case with replacement. Expression (G.2) ranges between 0 and 2μ .

In the case of without replacement, the sum is obviously to be divided by $N(N-1)$ as there are N deviations with the self. It is not difficult to see that the numerical value of the sum remains the same.

In order to make it serve as a measure of inequality, (G.2) can be divided by μ to produce what can be called coefficient of mean difference (CMD):

$$CMD = \frac{1}{N^2 \mu} \sum_{i=1}^N \sum_{j=1}^N |x_i - x_j| \quad (G.3)$$

CMD fulfils the idea of scale independence. However, the expression (G.3) ranges between 0 when everybody has the same income and $2[=2N/(N-1)]$ when only one person has all the income. In order to make it satisfy the interval (0,1), we can further divide it by 2. The result is Gini coefficient of concentration or Gini index of inequality:

$$G = \frac{1}{2N^2 \mu} \sum_{i=1}^N \sum_{j=1}^N |x_i - x_j| \quad (G.4)$$

or

$$G = \frac{1}{N^2 \mu} \sum_{i=1}^N \sum_{x_j \leq x_i} (x_i - x_j) \quad (G.5)$$

conceived as an aggregate of only positive differences, though normalized by the number of all differences and the mean income. Kendall and Stuart define this as 'one half of the average value of absolute differences between all pairs of incomes divided by the mean income'.

The index can also be defined in terms of population proportions and income shares. If the income-share of individual i is denoted by q_i , that is,

$$q_i = \frac{x_i}{N\mu}, \quad (G.6)$$

then the expression (G.4) can also be written as

$$G = \frac{1}{2N^2} \sum_{i=1}^N \sum_{j=1}^N |q_i - q_j| \quad (G.7)$$

In the case of a discrete distribution, each individual constitutes $(1/N)$ th of the population, that is,

$$p_i = \frac{1}{N}. \quad (G.8)$$

Therefore one can also write (G.7) as

$$G = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N |p_j q_i - p_i q_j| \quad (G.9)$$

An obvious question is: why not $|p_i q_i - p_j q_j|$ in the expression (G.9)? For understanding this, let us consider the Gini coefficient for the groups.

Let μ_r and μ_s denote mean incomes of r^{th} and s^{th} groups (say, families) respectively and $r, s=1, 2, \dots, g$. Then, Gini for the groups can be defined as

$$G = \frac{1}{2N^2\mu} \sum_{r=1}^g \sum_{s=1}^{Ng} |\mu_r - \mu_s| f_r f_s \quad (G.10)$$

where f_r and f_s are frequencies of the groups r and s respectively. This can obviously be written as

$$G = \frac{1}{2} \sum_{r=1}^g \sum_{s=1}^g \left| \frac{\mu_r}{\mu} - \frac{\mu_s}{\mu} \right| p_r p_s \quad (G.12)$$

where

$$p_r = \frac{f_r}{N} \text{ and } p_s = \frac{f_s}{N} \quad (G.12)$$

Now, let us note the share of total income with the group r :

$$q_r = \frac{\mu_r f_r}{N\mu} = p_r \frac{\mu_r}{\mu} \quad (G.13)$$

Then, the expression (G.12) can be written in either of the two ways (G.14) and (G.15)

$$G = \frac{1}{2} \sum_{r=1}^g \sum_{s=1}^g \left| \frac{q_r}{p_r} - \frac{q_s}{p_s} \right| \quad (G.14)$$

or

$$G = \frac{1}{2} \sum_{r=1}^g \sum_{s=1}^g |p_s q_r - p_r q_s| \quad (G.15)$$

It is easy to see that $g=N$ and $p_s=p_r=(1/N)$ when f_r and f_s are all equal to 1.

In statistics literature we emphasize frequency aspects, in economics literature we find it convenient, expression-wise, to treat each individual with single income though there is no bar for $x_i=x_j$.

11.3.2 Simple Computational Device

Two years after giving his index to terms of relative mean differences, Gini (1914) showed that the index is exactly equal to one minus twice the area under Lorenz curve (to be discussed later). That is,

$$G = 1 - 2\bar{A} \quad (LG.1)$$

where $\bar{A}$ is the area under the Lorenz curve, as shown in Fig. 11.1

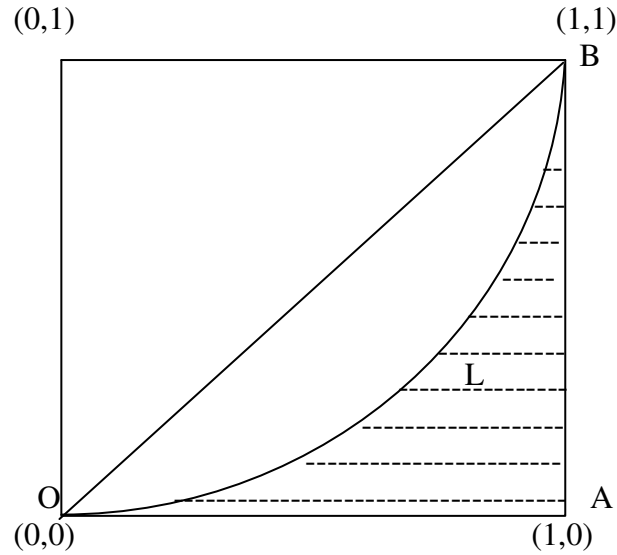


Fig. 11.1

However, normally, we do not specify and estimate a smooth relationship between Q and P . Instead, we obtain the curve by plotting cumulative proportions of people in classes and cumulative shares of their incomes, where classes are arranged according to increasing per capita income values:

$$\mu_1 \leq \mu_2 \leq \mu_3 \leq \dots \leq \mu_r \leq \dots \leq \mu_g. \quad (\text{LG.4})$$

strictly speaking, equality sign is useless in this presentation. (We have written it in deference to Theil (1967)). Plotting P_r and Q_r , we obtain the Fig. 11.2. We can see that the area below the Lorenz curve can be conceived as consisting of several trapeziums. A trapezium could be seen as consisting of a rectangle and a triangle. Summing the areas of all trapeziums (say g in number), we can get the area $\bar{A}$. Substituting it in (LG.1), we get the following expression for computing Gini coefficient G :

$$G = 1 - \sum_{r=1}^g (P_r - P_{r-1})(Q_r + Q_{r-1}) \quad (\text{LG.5})$$

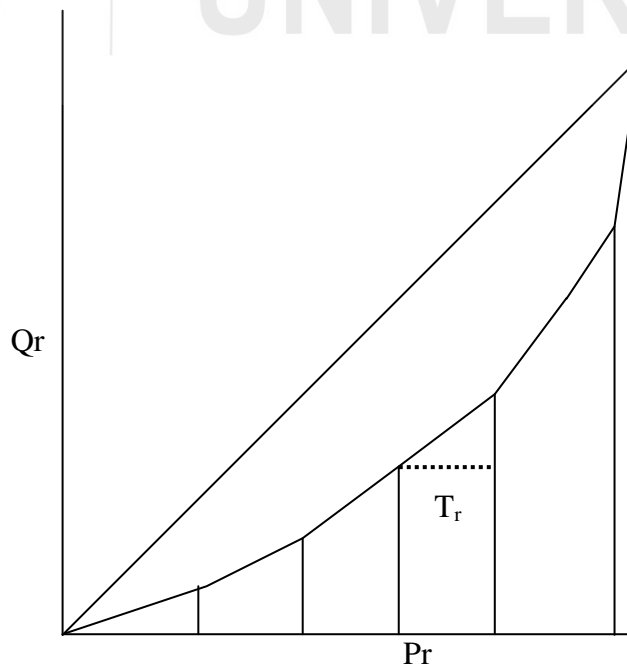


Fig. 11.2

1) Define Gini ratio.

.....

.....

.....

.....

.....

2) Show the difference in approach in defining Gini from other measures of dispersion.

.....

.....

.....

.....

.....

3) Write the expression for computing Gini.

.....

.....

.....

.....

.....

4) How can you compute Gini ratio?

.....

.....

.....

.....

.....

11.4 LORENZ CURVE

Lorenz curve is a powerful geometrical device to compare two situations of distribution with regard to their level of inequality. Devised some hundred years ago by Max O. Lorenz (1905) to measure concentration of wealth, it is still very widely used in empirical studies on inequality. The device can be used for comparing inequality of distribution of any measurable entity such as

income, wealth (land, capital), consumption, expenditure on an item (say, food or education), etc. The distribution may be over persons or households. But the device can also be used to measure inequality of tax collection or expenditure incurred by states or federal grants received by different states. We can compare pre-tax and post-tax distributions in order to study the efficacy of instrument of tax.

Lorenz (1905) studied a number of methods then in use to gauge the level of, or change in the level of, inequality. Most of these measures used fixed-income classes in data tabulation and made inter-temporal comparison, employed changes in percentage of recipients of class incomes in each of the fixed-income classes or movement of persons from one class to another and so on. Finding them unsatisfactory, he comes to the conclusion that changes in income and changes in population both have to be simultaneously taken into account and in a manner that 'fixed-ness' of income classes gets neutralized.

In fact, this measurement relates to comparison and in most cases, we are in a position to compare but there are situations of non-comparability. However, a few of inequality measures that are capable of numerical representation in terms of a scalar number, and therefore called summary measure, are found to be based on the Lorenz curve.

It may be pointed out that the curve was independently introduced by Gini (1914). It is therefore, quite often referred to as Lorenz-Gini curve as well. We shall, however, stick to more common usage and call it Lorenz Curve.

11.4.1 Geometrical Definition

The Lorenz curve of concentration of incomes is the relationship between the cumulative proportions of recipients, usually plotted on the abscissa, and the corresponding cumulative shares of total income with the recipients, usually plotted on the ordinate. If population proportions and income shares of class j are denoted by p_j and q_j and cumulative proportions and shares upto class i , by P_i and Q_i then

$$P_i = \sum_{j=1}^i p_j, \quad 1 \geq p_j \geq 0 \quad (\text{GD.1})$$

and

$$Q_i = \sum_{j=1}^i q_j, \quad 1 \geq q_j \geq 0 \quad (\text{GD.2})$$

The relationship between P_i and Q_i is given by the curve

$$Q_i = L(P_i), \quad 1 \geq P_i \geq 0, \quad 1 \geq Q_i \geq 0 \quad (\text{GD.3})$$

and the point on the curve by (P_i, Q_i) . Naturally, the first point is (0,0) and the last one on the curve, (1,1). It is also clear that $Q_i \leq P_i$ $i=1, 2, \dots, N-1$ if there are N classes of incomes. It means no point will make an angle of more than 45° with the abscissa at the origin. Then, one can be sure that the Lorenz curve lies in the lower triangle of Lorenz Box of the unit square. See Fig. 11.3 in which OLB shows the Lorenz curve (Fig. 11.3).

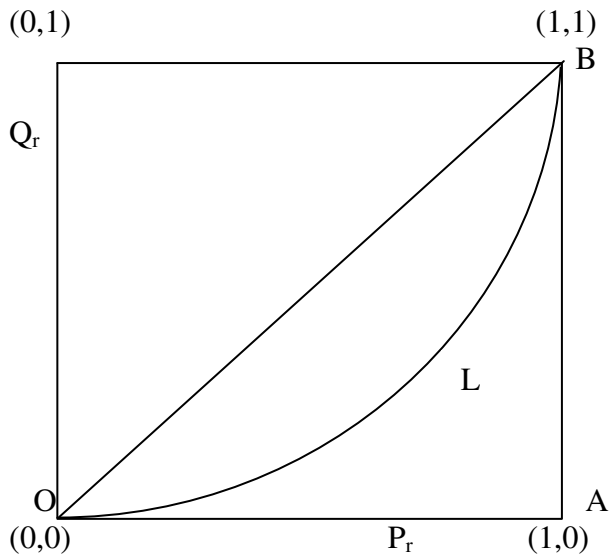


Fig. 11.3

11.4.2 Properties of the Lorenz Curve

Now it is easy to see that the extreme case of perfect equality is given by the diagonal OB which represents $P_i = Q_i$, $i = 1, 2, \dots, N$. The other extreme of perfect inequality will be given by a curve OAB. The diagonal OB is often designated as the egalitarian line or line of equality. The other diagonal CA is known as the alternative diagonal and is useful to study the symmetry of the curve. The triangle OAB with sharp kink of 90° at A can be said to be the line of perfect inequality. See Fig. 11.4.

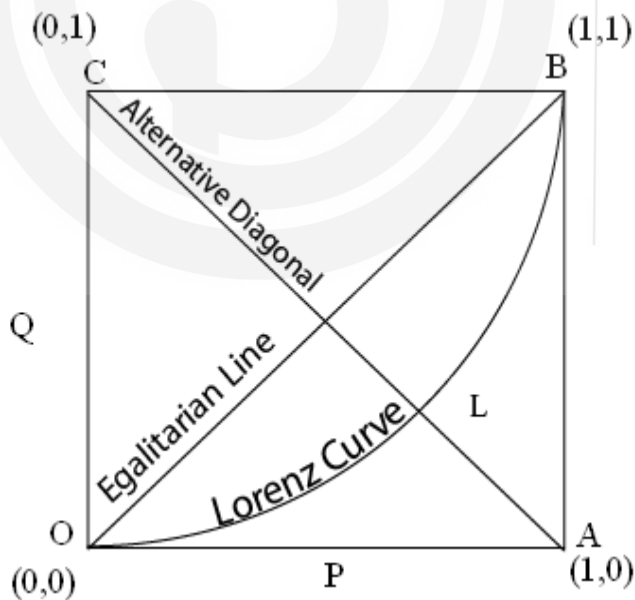


Fig. 11.4

We can finally note the following properties:

- i) $1 \geq p_i \geq 0; 1 \geq q_i \geq 0, i = 1, 2, \dots, N$
- ii) $1 \geq P_i \geq 0; 1 \geq Q_i \geq 0, i = 1, 2, \dots, N - 1$
- iii) $P_0 = Q_0 = 0; P_N = Q_N = 1$
- iv) $P_i \geq Q_i, i = 1, 2, \dots, N - 1$

By drawing a Lorenz Curve, we can know whether a given distribution is equal or unequal. We do not yet know how much unequal a given distribution is. When we draw two or more Lorenz Curves, we can compare the distributions as regards their levels of inequality. The curve closer to the diagonal of equality has lower level of inequality than the one away from it (Fig. 11.5). But we do not know yet the level of inequality. And even this comparison is possible only when the curves do not intersect (Fig. 11.6).

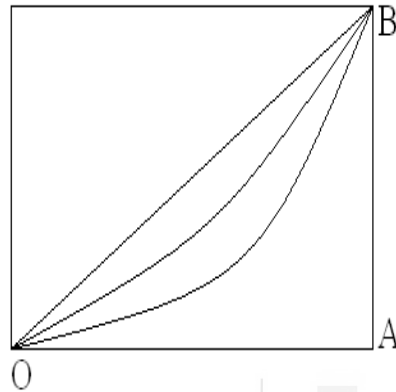


Fig. 11.5

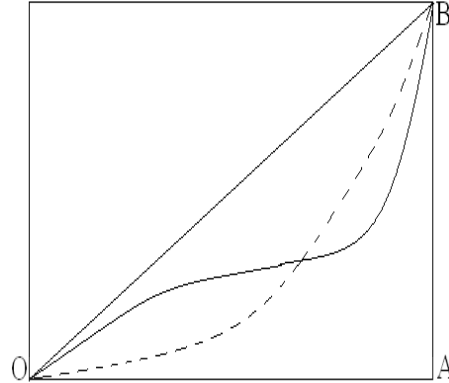


Fig. 11.6

However we can devise some measures, which are based on the Lorenz curve. In case the Lorenz curves intersect, reducing the distributions into single real number is the only option. So we shall discuss only two such proposals.

11.4.3 A Measure Based on Area

We have noted that if Lorenz curve coincides with the diagonal of equality, the inequality is nil and if Lorenz curve coincides with the two sides of the square, the inequality is full.

In the case of non-intersecting Lorenz curves, it is clear that the curve closer to the diagonal of equality will circumscribe smaller area between itself and the diagonal of equality than the one, which is farther. Which is what it should be. We can therefore devise a measure of inequality by dividing the area OLB (Fig. 11.4) by the area of triangle OAB, which is the maximum possible area between the diagonal of equality and Lorenz curve. As the area of OAB is $(1/2)$, the measure turns out to be twice the area between the diagonal of equality and the Lorenz curve. In other words, Lorenz coefficient of concentration (LCC) is:

$$LCC = \frac{\text{Area OLB}}{\Delta OAB} = 2 \text{Area OLB}$$

Since this turns out to be exactly equal to Gini coefficient, we are not elaborating it any further.

11.4.4 A Measure Based on Length

This is a measure proposed by Kakwani (1980). The length of the Lorenz curve, denoted by ℓ , cannot fall below $\sqrt{2}$, which is the length of the egalitarian line and cannot exceed 2, which is the sum of the lengths of the two arms of the lower triangle. In order to produce a measure with the minimum value 0 and the maximum value 1, following exercise can be suggested:

	Minimum	Actual	Maximum
Length of the Curve	$\sqrt{2}$	ℓ	2
Length of the Curve - $\sqrt{2}$	0	$\ell - \sqrt{2}$	$2 - \sqrt{2}$
$\frac{\text{Length of the Curve} - \sqrt{2}}{\text{Maximum length} - \sqrt{2}}$	0	$\frac{\ell - \sqrt{2}}{2 - \sqrt{2}}$	1

So this measure is clearly:

$$LK = (\ell - \sqrt{2}) / (2 - \sqrt{2})$$

In both the cases, one can draw actual graphs and actually measure the area and the length and calculate the indices for level of inequality. Those who wish to carry out a more sophisticated exercise will have to estimate smooth functions.

Check Your Progress 3

- 1) Enumerate the properties of Lorenz curve.

.....

.....

.....

.....

- 2) When will comparison between two Lorenz curve fail to compare inequality in two distribution?

.....

.....

.....

.....

- 3) What is the relationship between Lorenz curve and Gini coefficient.

.....

.....

.....

.....

- 4) What is Kakwani's measure of inequality, which is based on the Lorenz curve.

.....

.....

.....

.....

.....